



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-3)

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❖ Tokens

- A token is the smallest individual element of a C++ program that is meaningful to the compiler. Tokens act as a building blocks of a program.
- The following tokens are available in C++ keywords, identifiers, constants, variables, and operators described in this section are examples of tokens. Punctuation characters such as brackets ([]), braces ({ }), parentheses (()), and commas (,) are also tokens.

➤ Name of tokens

- ✓ Keywords
- ✓ Identifiers
- ✓ Constants
- ✓ Variables
- ✓ Operators
- ✓ Punctuation

❖ Variables

- In C++ a variable is a name of memory location used to store data. A variable is a location in your computer's memory in which you can store a value and from which you can later retrieve that value.
- The value stored in a variable can change during program execution.
- When you define a variable in C++, you must tell the compiler what kind of variable it is: an integer, a character, and so forth.
- When you run your program, it is loaded into RAM from the disk file. All variables are also created in RAM.

Declaration of Variable

Declaration will allocate memory for specified variable with garbage value.

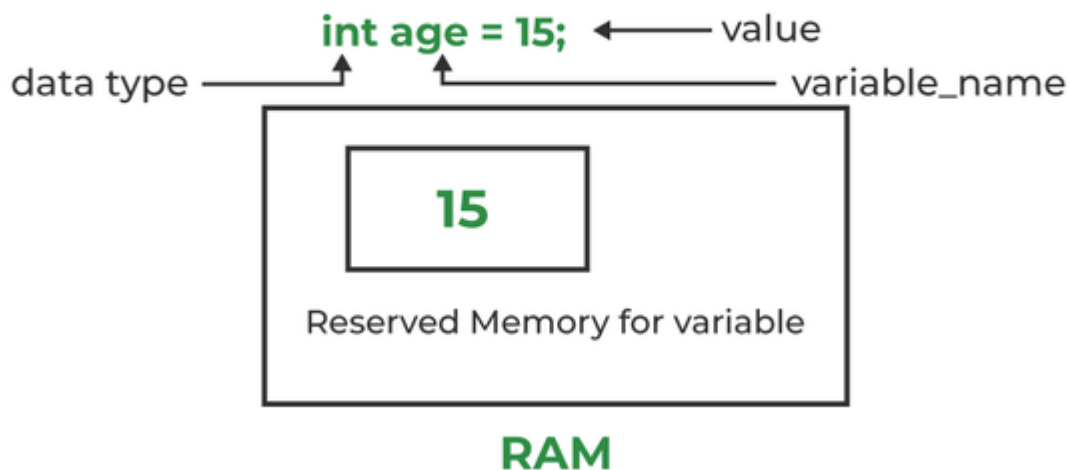
Syntax :

```
Data-Type    Variable-name;
```

Examples :

```
int a;  
float b;  
char c;
```

Variable in C++



❖ Initialization of Variable in Program

Initialization means assigning the first value to a variable at the time of declaration. Every new value will overrides the previous value.

```

1  #include<iostream>
2  using namespace std;
3  #include<conio.h>
4  int main()
5  {
6      int a=10, A =10; //Initialization of Variable
7      cout << a << endl << A <<endl;
8      a=75, A =22.5; //Initialization of Variable
9      cout << a << endl << A <<endl;
10     int hira=2026; //Initialization of Variable
11     cout << hira;
12     float f=5.5; //Initialization of Variable
13     cout << f <<endl;
14     char c='y'; //Initialization of Variable
15     cout << c << endl;
16     string bihar = "Hira"; //Initialization of Variable
17     cout << bihar;
18
19     getch();
20 }
21

```

C:\Users\hira\Desktop\z.exe

```

10
10
75
22
20265.5
y
Hira

```

Use this code to input any string with spaces.

```

hello.cpp x
#include<iostream>
#include<string>
#include<conio.h>
using namespace std;
int main()
{
    string s;
    cout << " Enter your name : ";
    getline(cin, s);
    cout << " Thank You !" << endl <<endl;
    cout << " Your name is : " << s;
    getch();
}

```

C:\Users\hira\Desktop\hello.exe

```

Enter your name : Hira kumar
Thank You !

```

```

Your name is : Hira kumar

```

Q. Write a program to print the user's name, age, and DOB using user input & what answer of 5/2 in c++.

```

1  #include<iostream>
2  using namespace std;
3  #include<conio.h>
4  int main()
5  {
6      int a,d; float f; // Initialization of Variable
7      char c;          // Initialization of Variable
8      string s;        // Initialization of Variable
9      cout << " What is your name = ";
10     cin >> s;
11     cout << s << endl;;
12     cout << "What is your age = ";
13     cin >> a;
14     cout <<a << endl;
15     cout << "What is your DOB = ";
16     cin >> d;
17     cout <<d <<"\n";
18     cout << "What answer of 5/2 = "; cin >>f;
19     cout << f;
20     getch();
21 }
22

```

C:\Users\hira\Desktop\z.exe

```

What is your name = hira
hira
What is your age = 99
99
What is your DOB = 01011990
1011990
What answer of 5/2 = 2.5
2.5

```

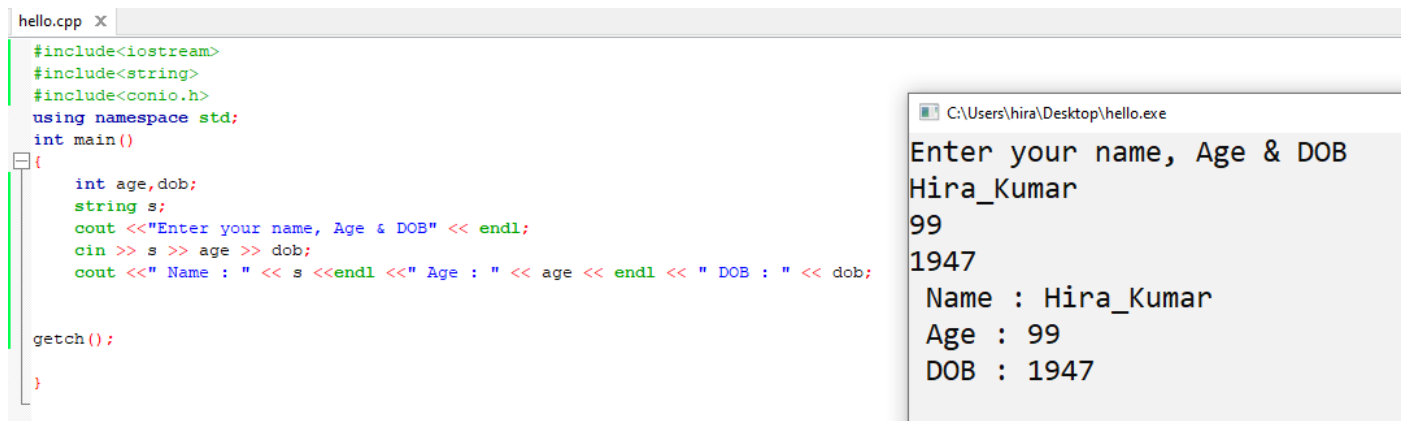
HIRA KUMAR

6. “Optimize or “Minimize” above c++ program. (Write a program to print name, age, DOB of user & what answer of 5/2 in c++.)

OR

Reduce the number of lines while keeping the same logic.

Q6 Answer

The image shows a code editor window titled 'hello.cpp' with the following C++ code:

```
#include<iostream>
#include<string>
#include<conio.h>
using namespace std;
int main()
{
    int age,dob;
    string s;
    cout <<"Enter your name, Age & DOB" << endl;
    cin >> s >> age >> dob;
    cout <<" Name : " << s <<endl <<" Age : " << age << endl << " DOB : " << dob;

    getch();
}
```

Next to the code editor is a screenshot of the program's execution. The window title is 'C:\Users\hira\Desktop\hello.exe'. The output is:

```
Enter your name, Age & DOB
Hira_Kumar
99
1947
Name : Hira_Kumar
Age : 99
DOB : 1947
```

❖ C++ Identifiers

In C++ programming language, identifiers are the unique names assigned to variables, functions, classes, structs, or other entities within the program.

➤ Rule = How to write Variable_name / Identifiers

- It must begin with alphabet or underscore. (e.g – a2, Hira, _2, bihar_sharif,)
- No other symbol is allowed (e.g – a\$, a.b, x-1,...)
- 1st variable name cannot start with a digit (e.g – 2a, 8x,...)
- C++ has reserved keywords that cannot be used as identifiers.
- Identifier must be unique in its namespace.

Valid Identifiers	Invalid Identifiers
- Hira; - A; - num; - bihar_sharif; - a2; - sum;	1hira (start with digit) - bihar sharif (space not allow) - goto (keyword) 2a; (start with digit) - Bihar@123 (special symbol not allow)

❖ Keywords

- It is also called Pre-define words / Reserved word /type of token
- Keywords (reserved words) have special meanings to the C++ compiler and are always written or typed in lower cases.
- Keywords are words that the language uses for a special purpose, such as void, int, public, etc.
- It can't be used for a variable name or function name or any other identifiers.
- The total count of keywords is 95, in which 32 keywords present in C language.
- Categorization of c++ keywords 

Examples of Keywords:

asm	double	new	<u>switch</u>
auto	else	operator	template
break	enum	private	this
case	extern	protected	throw
catch	float	public	try
char	for	register	typedef
class	friend	return	union
const	goto	short	unsigned
continue	<u>if</u>	signed	virtual
default	inline	sizeof	void
delete	int	static	volatile
do	long	struct	while

❖ Constants

- Constants (often referred to as Literals) are data items that never change their value during the execution of the program.

Type of Constants

The following types of constants are available in C++.

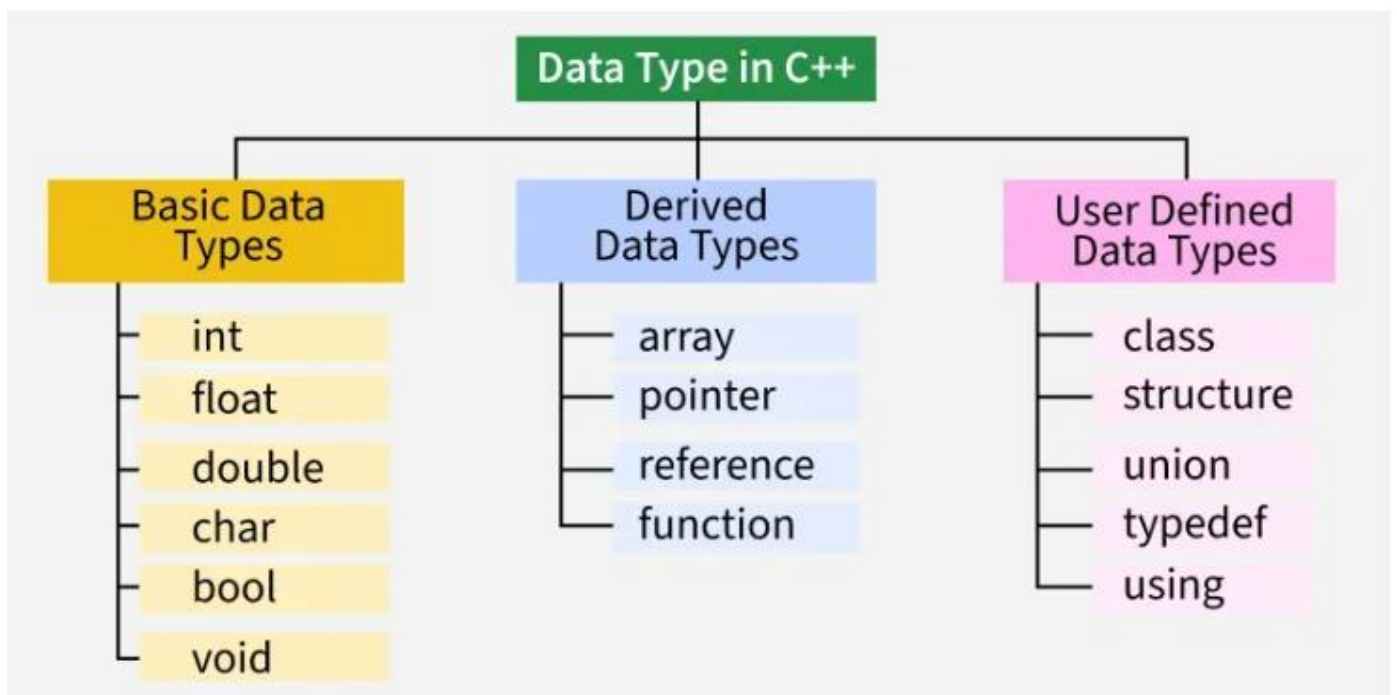
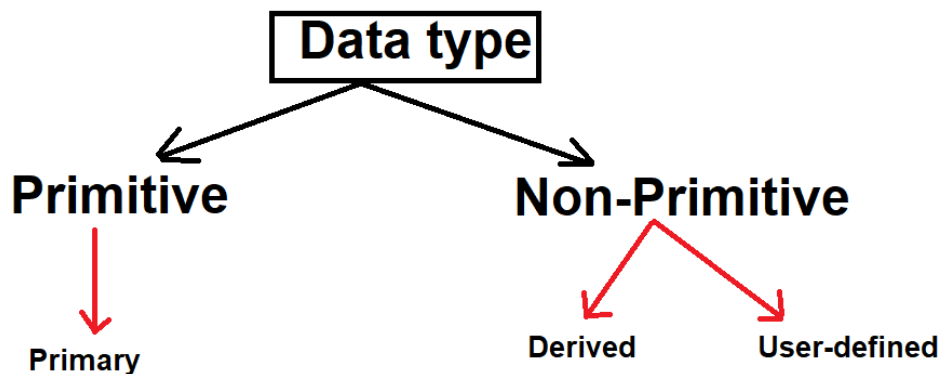
- ✓ Integer Constants
- ✓ Character Constants
- ✓ Floating Point Constants
- ✓ Strings Constants
- **Integer constant**-Consists of the digits.
 - ✓ Example :- 10, -555, 10A(hexa form), 22, -9,
- **Float constant**-Consists of the digits with decimal.
 - ✓ Example :- 3.0, -7.5, -0.067, 45.55, 2.5e3,
- **Character constant**-single character enclose with in a part of single quote.
 - ✓ Example :- 'A', '9', '#',
- **String constant**- Collection is sequence of characters enclosing in double quotes.
 - ✓ Example :- "Hira", "New Delhi",

❖ Data Type

- The data type determines the kind of information that may be stored in the variable. Variables are classified according to their data type.
- Whenever a variable is defined in C++, the compiler allocates memory for that variable based on the data type with which it is declared.
- The programmer can select the data type appropriate to the needs of the application. Data types specify the size and types of values to be stored.

C++ supports the following data types:

1. Primary or Built-in or Fundamental/Basic data type
2. Derived data types
3. User-defined data types



❖ Size of Primitive Data type

(64 bit compilers – 2025-2026)

Data Type	Size (byte)	Use	Range
int	4	To store integers numbers	-2147483648 to 2147483647
long	4	For large integers numbers	-2147483648 to 2147483647
long long	8	For very large integers numbers	-9223372036854775808 to 9223372036854775807

float	4	For decimal numbers	-3.4×10^{38} to 3.4×10^{38}
double	8	For high precision decimal numbers	$\pm 1.7 \times 10^{308}$
long double	8	For extra high precision decimal numbers (some compiler 12/16)	$\pm 1.1 \times 10^{4932}$
char	1	To store a single character	'A', 'a', '0',
bool	1	To store true/false values	True/false

```

1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      // size in byte
7      cout << "Size of Int = " << sizeof(int) << endl;
8      cout << "Size of Float = " << sizeof(float) << endl;
9      cout << "Size of Char = " << sizeof(char) << endl;
10     cout << "Size of Long = " << sizeof(long) << endl;
11     cout << "Size of Long long = " << sizeof(long long) << endl;
12     cout << "Size of Double = " << sizeof(double) << endl;
13     cout << "Size of Bool = " << sizeof(bool) << endl;
14     cout << "Size of void = " << sizeof(void) << endl;
15     getch();
16     return 0;
17 }

```

C:\Users\hira\Desktop\z.exe

```

Size of Int = 4
Size of Float = 4
Size of Char = 1
Size of Long = 4
Size of Long long = 8
Size of Double = 8
Size of Bool = 1
Size of void = 1

```

❖ C++ Modifiers

- Modifiers are used with primary data types like int, float, char to control size, sign, or behavior.

Data type modifiers available in C++ are:

- ✓ Signed → Can store positive & negative values
- ✓ Unsigned → Can store only positive values
- ✓ Short → Reduces the size of integer type
- ✓ Long → increases the size of integer type

Modifiers in C++



Old 16bit /Turbo C++ sizes

Type	Size (bits)	Size (bytes)	Range
char	8	1	-128 to 127
unsigned char	8	1	0 to 255
int	16	2	-2^{15} to $2^{15}-1$
unsigned int	16	2	0 to $2^{16}-1$
short int	8	1	-128 to 127
unsigned short int	8	1	0 to 255
long int	32	4	-2^{31} to $2^{31}-1$
unsigned long int	32	4	0 to $2^{32}-1$
float	32	4	3.4E-38 to 3.4E+38
double	64	8	1.7E-308 to 1.7E+308
long double	80	10	3.4E-4932 to 1.1E+4932

Revision 1st year

❖ What is Sign & unsign integer

- **Signed integer** : A signed integer can store both positive(+ve) and negative (-ve) number. The leftmost bit (MSB) is used as the sign bit.

0 = positive

1 = negative

Example : 4 bit signed integer

1 = negative	0 = Positive
	
MSB	
1000 = -8	0000 = 0
1001 = -7	0001 = 1
1010 = -6	0010 = 2
1011 = -5	0011 = 3
1100 = -4	0100 = 4
1101 = -3	0101 = 5
1110 = -2	0110 = 6
1111 = -1	0111 = 7

- **Unsigned Integer** : An unsigned integer can store only non-negative (0 and positive) numbers.

Example : 4 bit unsigned binary

0 = Positive	0 = Positive
	
0000 = 0	1000 = 8
0001 = 1	1001 = 9
0010 = 2	1010 = 10
0011 = 3	1011 = 11
0100 = 4	1100 = 12
0101 = 5	1101 = 13
0110 = 6	1110 = 14
0111 = 7	1111 = 15

hello.cpp X

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int a = 2147483647; // signed integer use
    cout << "value of a = " << a << endl;
    int b = 2147483650; // over limit
    cout << "value of b = " << b << endl;
    unsigned int c = 2147483650; // unsigned used
    cout << "value of c = " << c << endl;

    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
value of a = 2147483647
value of b = -2147483646
value of c = 2147483650
```

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